Shubham

+91 7988358944 shubham.741123@gmail.com shubham774 Shubham-741 shubham_778 shubham7711

Education

Bennett University **2021 – 2025**
B.Tech in Computer Science and Engineering (Specialization: Data Science) *CGPA: **9.63***

Delhi Public School, Rohtak **2019 – 2020**
Class 12, Central Board of Secondary Education *Percentage: **91.66%***

Scholars Global School, Bahadurgarh **2016 – 2017**
Class 10, Central Board of Secondary Education *Percentage: **95%***

Experience

Samsung R&D Institute India, Delhi **Feb 2026 – Present**
Software Engineer – Design & Operations of Cloud Platforms

- Designed and implemented a distributed backend platform for Samsung TV+'s **QC Automation** using **Python, Flask, Redis, MQTT, MySQL, and REST APIs**, automating content validation for **4,000+** Live TV channels and VOD assets, processing **10,000+ validation jobs/day** while reducing manual effort by **80%**.
- Built cloud-native, event-driven microservices for the **QC Automation Platform** using **AWS EC2, Lambda, S3, SNS, RDS, CloudFront, IAM**, **Redis**, Linux workers, and **FFmpeg** for asynchronous, fault-tolerant media processing.
- Contributed to the **Samsung Stream Monitoring System** by building distributed monitoring pipelines for concurrent analysis of **4,000+ live channels**, validating **HLS (M3U8)** playlists, MPEG-TS segments, subtitles, bitrate variants, and stream availability, achieving **92% anomaly detection accuracy** while reducing incident detection time by **60%**.
- Built observability pipelines using **Python, Datadog, CASCADA, and MySQL** to analyze streaming telemetry across **4,000+** channels, enabling automated release validation, production monitoring, and rapid incident diagnosis.

Samsung R&D Institute India, Delhi **Aug 2025 – Feb 2026**
Software Engineer Trainee

- Developed scalable metadata ingestion pipelines using **Python, PostgreSQL, SQL, and AWS** to process high-volume Samsung TV+ content feeds supporting content discovery and delivery services.
- Contributed to Samsung TV+'s **Click-to-Search (CTS)** by developing Computer Vision-based image retrieval models using **CLIP** and **Vision Transformer (ViT)** on **500K+ image snippets**, achieving **95% retrieval accuracy**. Built optimized **PostgreSQL** indexing for low-latency search across actors, directors, filmography, shows, and scene descriptions.

Siemens EDA **Jul 2024 – Jan 2025**
Software Engineer Intern

- Developed **CVE Checker Tool 4.0** using **Python, Django, and NVD APIs**, maintaining a continuously synchronized local CVE repository for low-latency vulnerability lookup and automated security analysis.
- Engineered Linux kernel analysis and patch verification pipelines to identify unpatched vulnerabilities, Dockerized the **CVE Checker Tool 4.0**, and automated HTML/Excel report generation for deployment in Siemens EDA's security framework.

Projects

Image Forgery Detection System using CNN | *Python, TensorFlow, Keras, CASIA2 Dataset* | GitHub

- Developed a specialized lightweight CNN model for binary classification of authentic vs. forged images, achieving **94.2% accuracy** on the CASIA2 dataset.
- Research paper **presented and accepted at IEEE Madhya Pradesh Section Conference 2025.**

Adaptive Traffic Control System | *Python, YOLOv3, OpenCV* | GitHub

- Engineered an adaptive traffic signal model using **YOLOv3-based vehicle detection**, lane segmentation, and weighted density computation to dynamically optimize signal timings.
- Validated on real-world traffic datasets, achieving **90.8% detection accuracy** in optimized traffic flow; research published in **2024 4th (CISCT).** [**IEEE Xplore Link**].

Cab Booking and Management System | *Node.js, JavaScript, MySQL, Dijkstra's Algorithm, RESTful APIs* | GitHub

- Developed a real-time cab booking backend using **Node.js** and **RESTful APIs**, integrating **Dijkstra's algorithm** for efficient shortest-path and fare computation.
- Implemented MySQL as the relational data store, enabling driver assignment, route-based fare calculation, and dynamic booking operations with optimal latency.

Technical Skills

Languages & Frameworks: Python, C++, Java, Django, Flask, FastAPI
Backend & Systems: REST APIs, Object-Oriented Programming (OOP), System Design, Operating Systems, Linux/Windows Internals, Computer Networking, AI/ML, LLMs, Agentic AI, Data Structures and Algorithms
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, DynamoDB, SQS, CloudFront), Redis, Git, Docker, Kubernetes
Databases: MySQL, PostgreSQL, MongoDB

Achievements

- **Qualified GATE CSE 2026**.
- Top 4% of the class in B.Tech (CSE). **Dean's List Award for Academic Excellence**.
- **Knight** on LeetCode (Rating: **2017**) | **5 Star** on HackerRank | **4 Star** on CodeChef (Rating: **1825**)